

COURSE CODE/TITLE:	CPE212 - Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	07/30/2026	DATE SUBMITTED:	07/30/2026
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ACTIVITY NO. 2.1 SSH Key-Based Authentication and GIT Setup

1. PROCEDURE OUTPUT

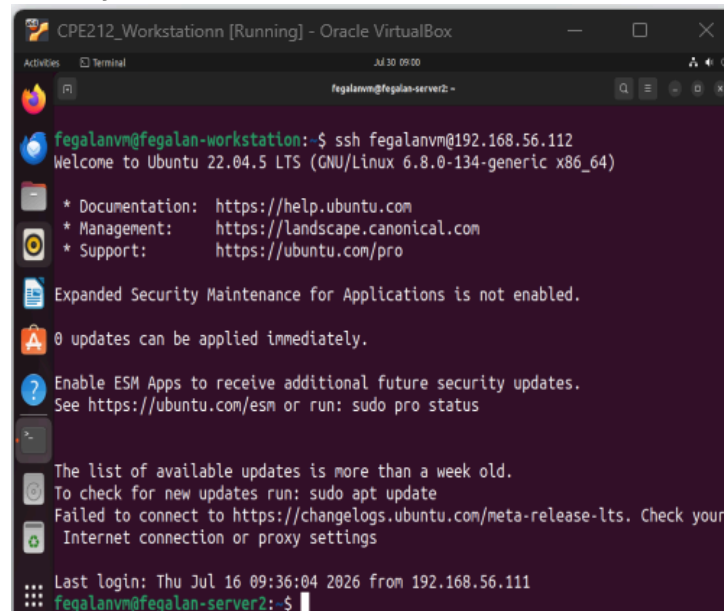
Task 1: Create an SSH Key Pair for User Authentication

```

The key's randomart image is:
+---[RSA 4096]---+
|o.+..o          |
|+o ... ..       |
|oo... ..o       |
|ooo+..+.        |
|.+=+o+o S       |
|.+.o.=E+.       |
|. .o = o        |
| oB.+ +         |
| oB0.o.o        |
+---[SHA256]-----+
fegalanvm@fegalan-workstation:~$ ls -la .ssh
total 24
drwx----- 2 fegalanvm fegalanvm 4096 Jul 30 08:42 .
drwxr-x--- 15 fegalanvm fegalanvm 4096 Jul 16 09:34 ..
-rw----- 1 fegalanvm fegalanvm 3401 Jul 30 08:42 id_rsa
-rw-r--r-- 1 fegalanvm fegalanvm 755 Jul 30 08:42 id_rsa.pub
-rw----- 1 fegalanvm fegalanvm 2240 Jul 16 09:50 known_hosts
-rw----- 1 fegalanvm fegalanvm 1120 Jul 16 09:37 known_hosts.old

```

Task 2: Copying the Public Key to the remote servers



```

CPE212_Workstationn [Running] - Oracle VirtualBox
fegalanvm@fegalan-server2:~$ ssh fegalanvm@192.168.56.112
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your
Internet connection or proxy settings

Last login: Thu Jul 16 09:36:04 2026 from 192.168.56.111
fegalanvm@fegalan-server2:~$

```

```
fegalanvm@fegalan-workstation:~$ ssh fegalanvm@192.168.56.113
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings

Last login: Thu Jul 16 09:50:00 2026 from 192.168.56.111
fegalanvm@fegalan-server1:~$
```

On the local machine, verify that you can SSH with Server 1 and Server 2. What did you notice? Did the connection ask for a password? If not, why? It did not ask for a password. This is because instead of typing passwords for logins, it uses keys or more specifically the ssh-copy-id sends its public host key to the remote server and saves it at the authorized_key and when trying to create a connection it sends an encrypted message that the remote decrypts and sends a message back that grants permission.

Reflections: Answer the following:

- 1. How will you describe the ssh-program? What does it do?** It creates a secure connection for remote logins to remote users using a local machine allowing the local machine to use the remote user without needing to be actively in front of the remote user. It allows for remote communication and type CLI commands through it.
- 2. How do you know that you already installed the public key to the remote servers?** You can check the remote server using `cat ~/.ssh/authorized_keys` to check for your public key.

Part 2: Discussion

```
fegalanvm@fegalan-workstation: ~
Get:1 http://ph.archive.ubuntu.com/ubuntu jammy/main amd64 liberror-perl all 0.1
7029-1 [26.5 kB]
Get:2 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 git-man all 1
:2.34.1-1ubuntu1.17 [954 kB]
Get:3 http://ph.archive.ubuntu.com/ubuntu jammy-updates/main amd64 git amd64 1:2
.34.1-1ubuntu1.17 [3,174 kB]
Fetched 4,154 kB in 7s (637 kB/s)
Selecting previously unselected package liberror-perl.
(Reading database ... 205465 files and directories currently installed.)
Preparing to unpack .../liberror-perl_0.17029-1_all.deb ...
Unpacking liberror-perl (0.17029-1) ...
Selecting previously unselected package git-man.
Preparing to unpack .../git-man_1%3a2.34.1-1ubuntu1.17_all.deb ...
Unpacking git-man (1:2.34.1-1ubuntu1.17) ...
Selecting previously unselected package git.
Preparing to unpack .../git_1%3a2.34.1-1ubuntu1.17_and64.deb ...
Unpacking git (1:2.34.1-1ubuntu1.17) ...
Setting up liberror-perl (0.17029-1) ...
Setting up git-man (1:2.34.1-1ubuntu1.17) ...
Setting up git (1:2.34.1-1ubuntu1.17) ...
Processing triggers for man-db (2.10.2-1) ...
fegalanvm@fegalan-workstation:~$ which git
/usr/bin/git
fegalanvm@fegalan-workstation:~$
```

CPE212 Workstation [Running] - Oracle VirtualBox

Activities Firefox Web Browser Jul 30 09:55

SSH and GPG keys x +

← → ↻ 🔒 # https://github.com/settings/keys

Settings

christopherfegalan (christopherfegalan)
Your personal account

- Public profile
- Account
- Appearance
- Accessibility
- Notifications
- Access
- Billing and licensing
- Emails
- Password and authentication
- Sessions
- SSH and GPG keys
- Credentials
- Organizations
- Enterprises
- Moderation

SSH keys

New SSH key

This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.

Authentication keys

Key Name	SHA256	Added	Used	Permissions	Action
CPE212	SHA256:s3eUyKqLQ77zp08e6nV3Q5owb/rW1C8YsQrDcCF7Z8	Added on Jul 30, 2026	Never used	Read/write	Delete

Check out our guide to [connecting to GitHub using SSH keys](#) or troubleshoot [common SSH problems](#).

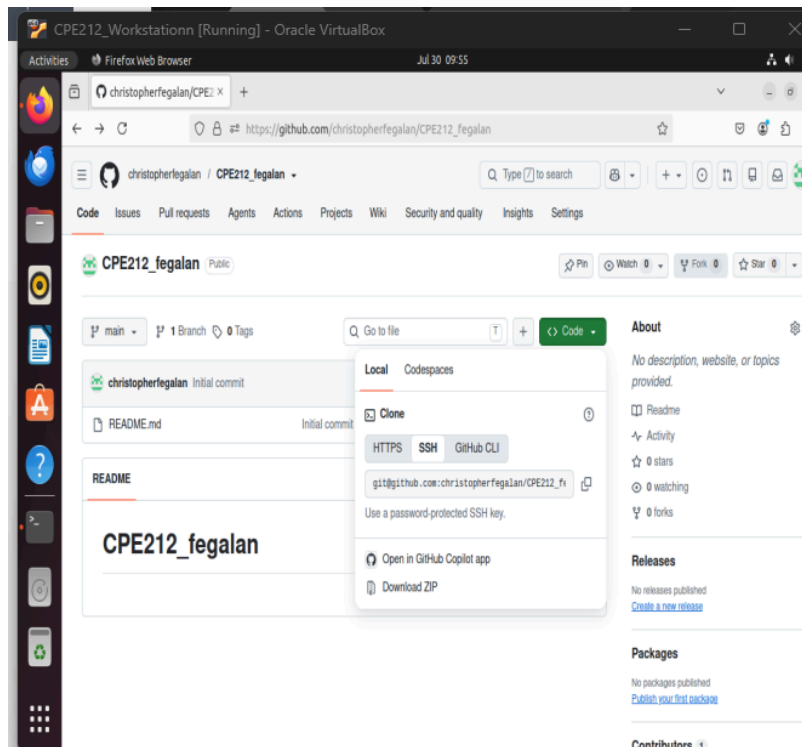
GPG keys

New GPG key

There are no GPG keys associated with your account.

Learn how to [generate a GPG key](#) and add it to your account.

Vigilant mode



```
fegalanvn@fegalan-workstation:~$ git clone git@github.com:christopherfegalan/CPE212_fegalan.git
Cloning into 'CPE212_fegalan'...
The authenticity of host 'github.com (20.205.243.166)' can't be established.
ED25519 key fingerprint is SHA256:+DiY3wvvV6TuJJhbpZisF/zLDA0zPMSvHdkr4UvC0qu.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'github.com' (ED25519) to the list of known hosts.
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
```

```
fegalanvn@fegalan-workstation:~$ ls
CPE212_fegalan  Documents  Music      Public  Templates
Desktop         Downloads  Pictures   snap    Videos
fegalanvn@fegalan-workstation:~$ cd CPE212_fegalan
fegalanvn@fegalan-workstation:~/CPE212_fegalan$ nano README.md
fegalanvn@fegalan-workstation:~/CPE212_fegalan$ git status
On branch main
Your branch is up to date with 'origin/main'.

changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

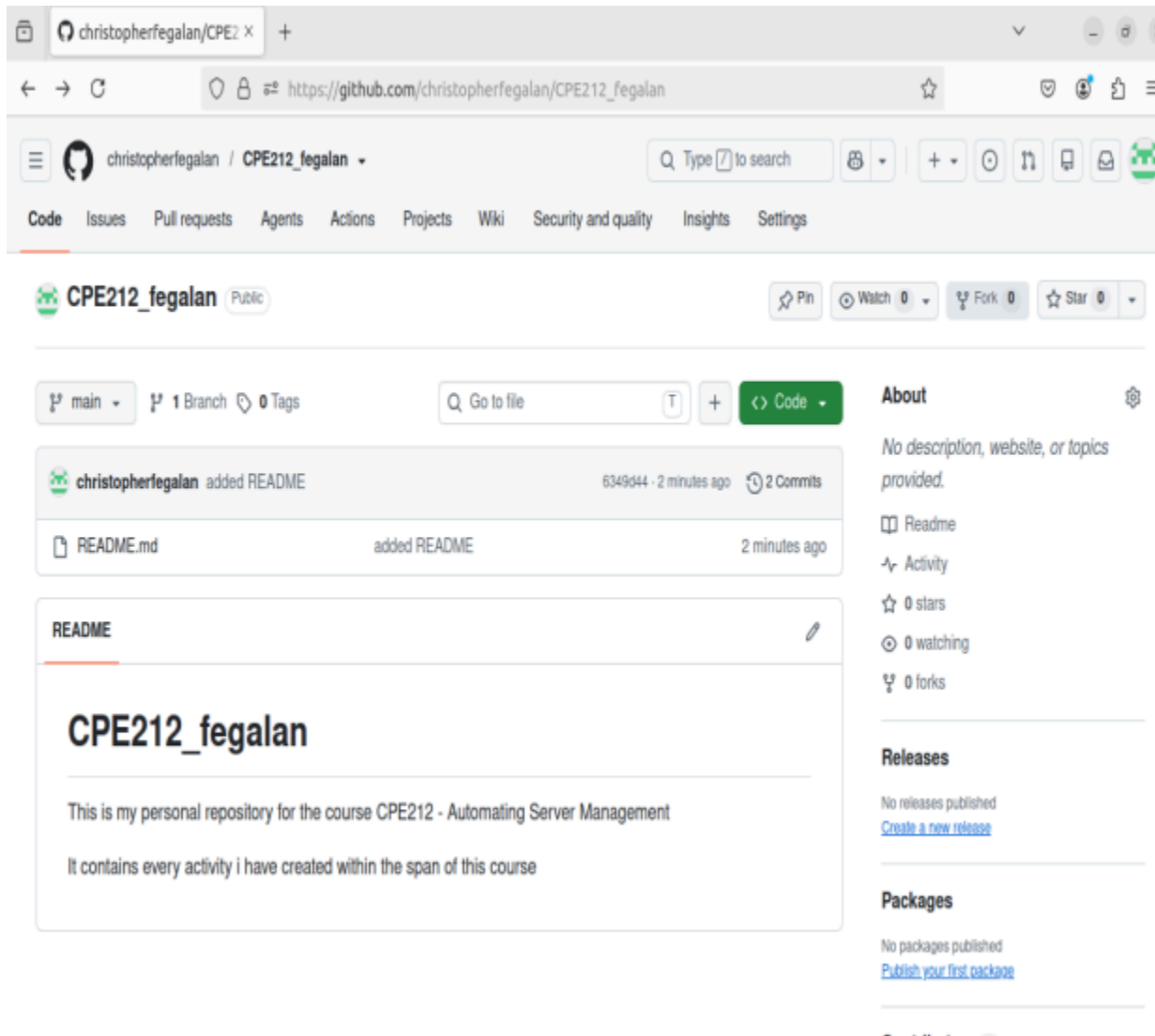
no changes added to commit (use "git add" and/or "git commit -a")
```

```
fegalanvn@fegalan-workstation:~/CPE212_fegalan$ git add README.md
fegalanvn@fegalan-workstation:~/CPE212_fegalan$ git commit -m "added README"
[main 6349d44] added README
1 file changed, 7 insertions(+), 1 deletion(-)
```

```

fegalanvm@fegalan-workstation:~/CPE212_fegalan$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 9 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 388 bytes | 388.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To github.com:christopherfegalan/CPE212_fegalan.git
736e0c5..6349d44  main -> main

```



The screenshot shows a web browser displaying the GitHub repository page for 'CPE212_fegalan' by user 'christopherfegalan'. The repository is public and has 0 stars, 0 forks, and 0 watches. The repository contains a single file named 'README.md' which was added 2 minutes ago. The README content is as follows:

CPE212_fegalan

This is my personal repository for the course CPE212 - Automating Server Management

It contains every activity i have created within the span of this course

The right sidebar shows the repository's metadata and links to the README, Activity, Stars, Watching, and Forks sections. The 'About' section is currently empty, and the 'Releases' and 'Packages' sections also show no published content.

Conclusions/Learnings:

- Learning to set up an ssh connection through github was so useful. This really helped me understand how useful ssh is. From setting up the ssh key, cloning the repo to my local machine and editing the cloned repo in my local machine and using git commands to push it back to the github repository were all very helpful to me in understanding how everything works and how git and github is used for project management.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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